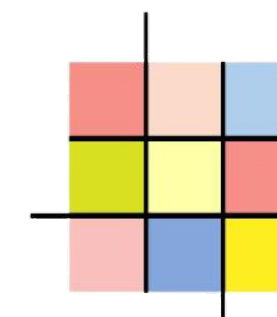


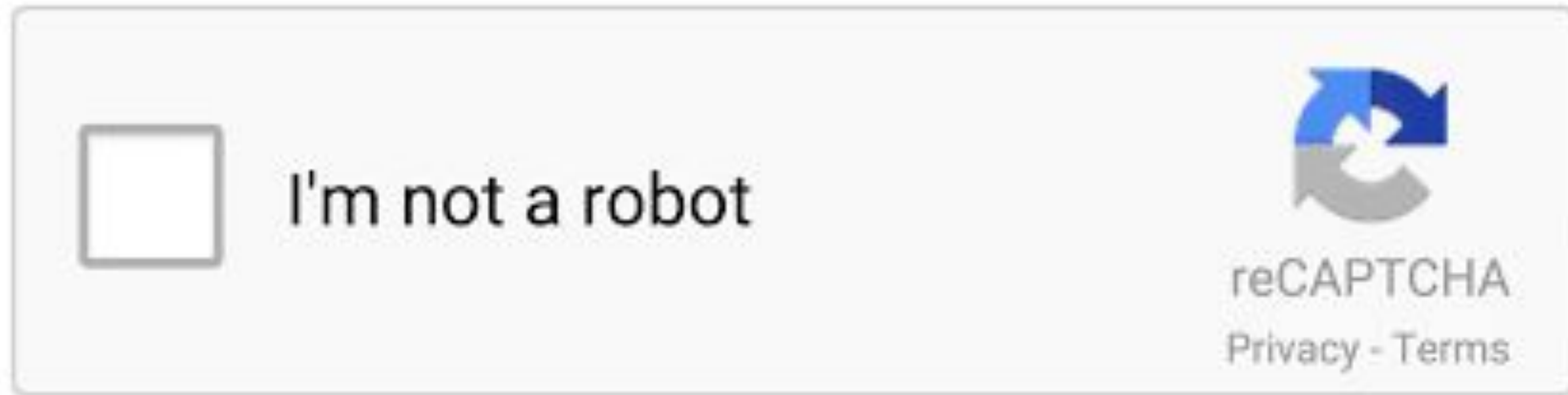
To what extent can the performance of the K-Means clustering algorithm be optimized on the MNIST dataset using different parameter settings in scikit-learn?

Clustering Algorithm

Research at FIS and Beyond, Feb 2026

Ziyou (Zoe) Zhang



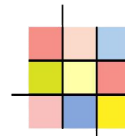


CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart)



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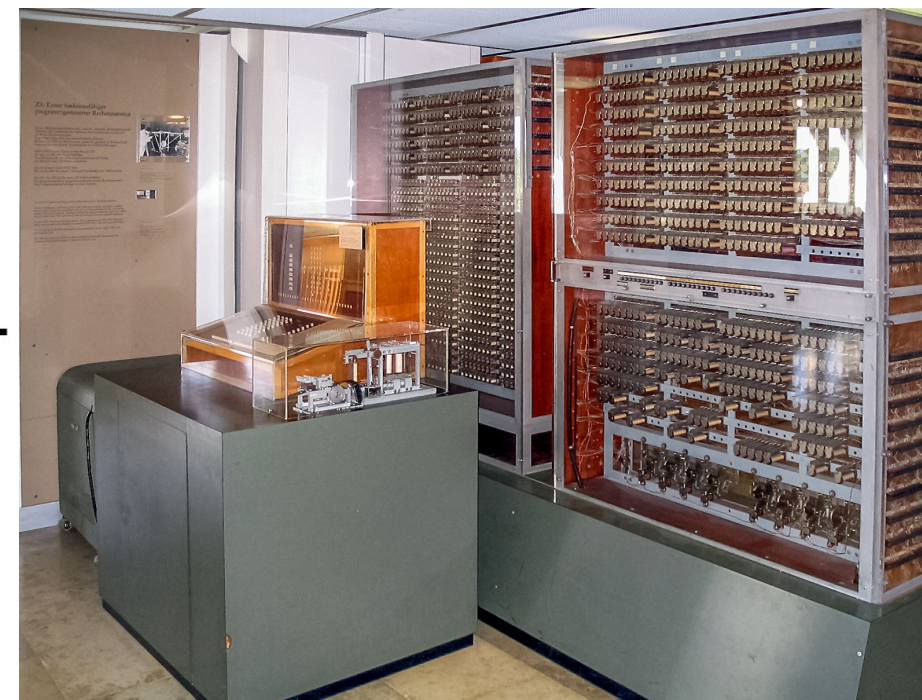
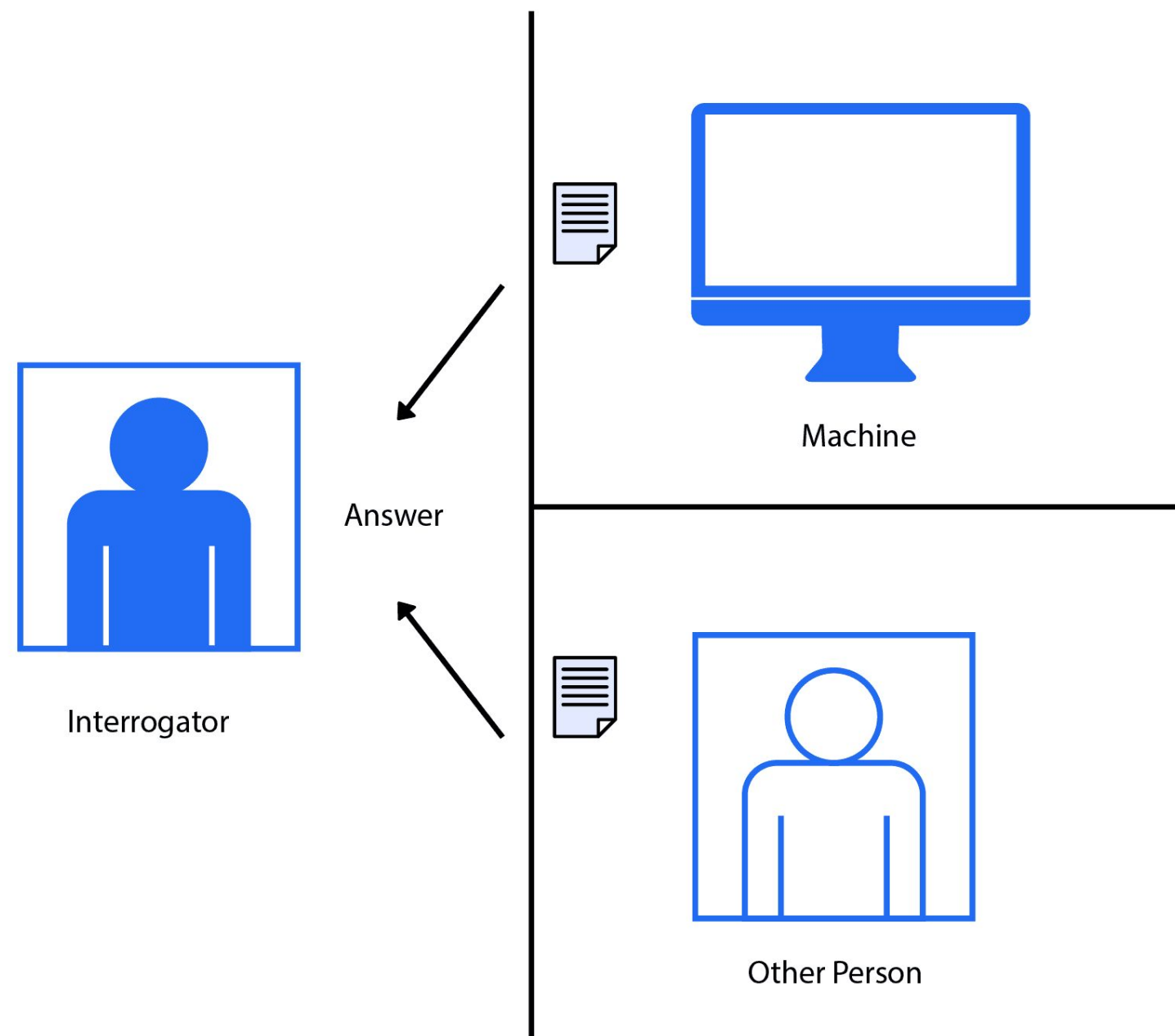


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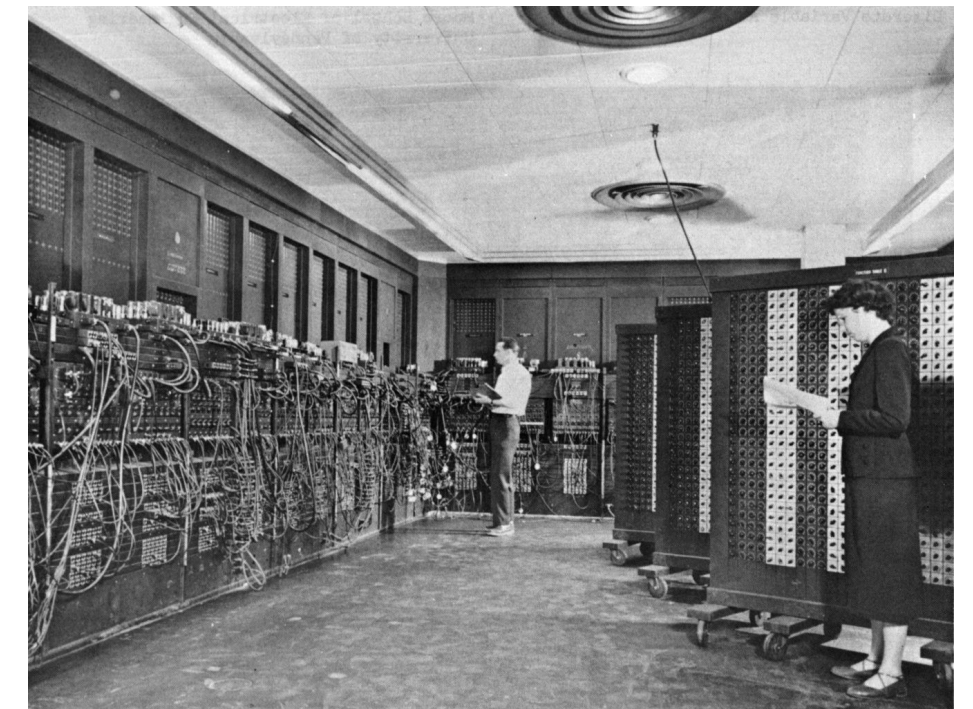
Turing test

Developed by Alan Turing in
1949, also called the imitation
game

Turing Test



Z3, 1941



ENIAC, 1945



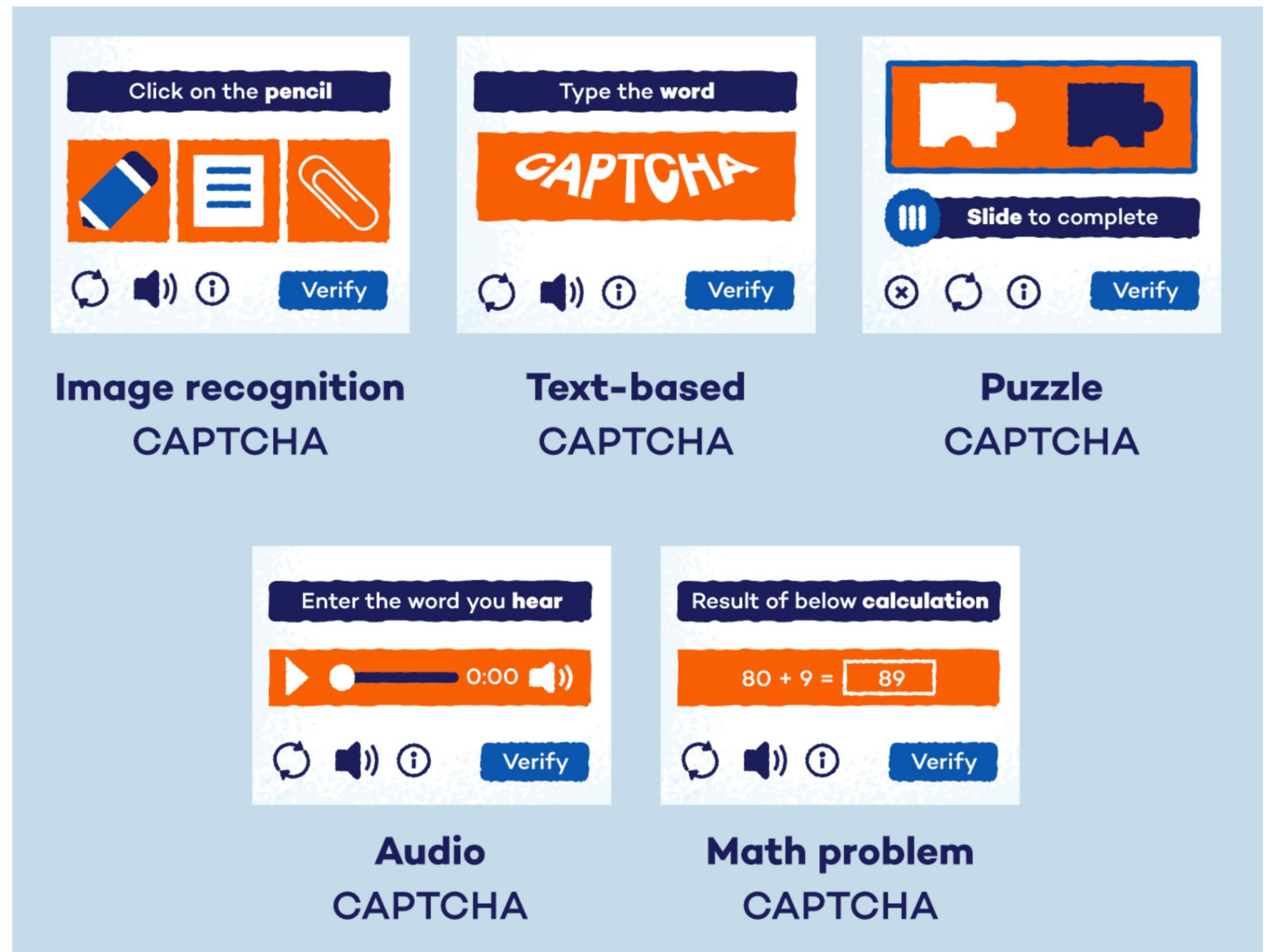
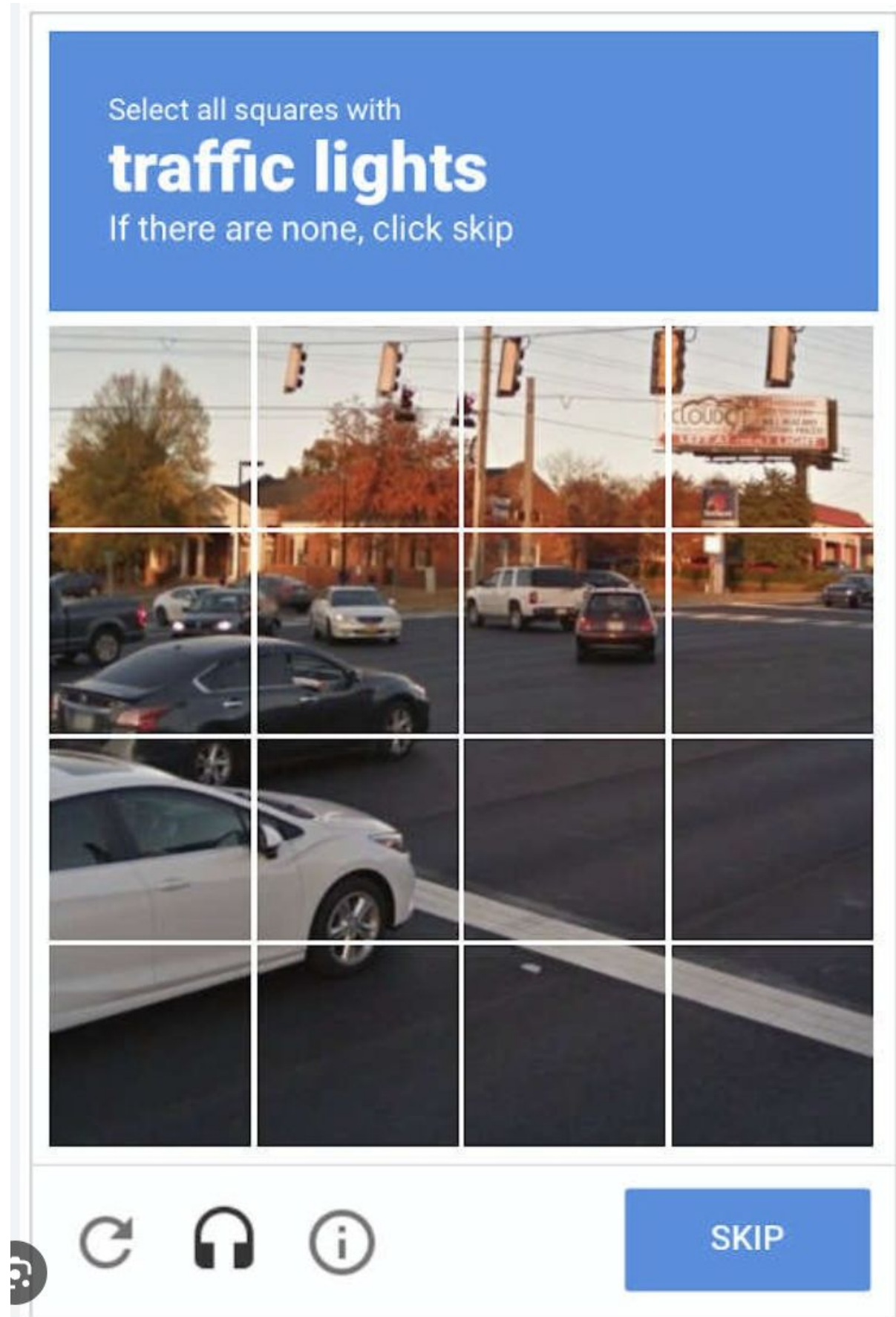
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CAPTCHAs



<https://www.pandasecurity.com/en/mediacenter/what-is-captcha/>

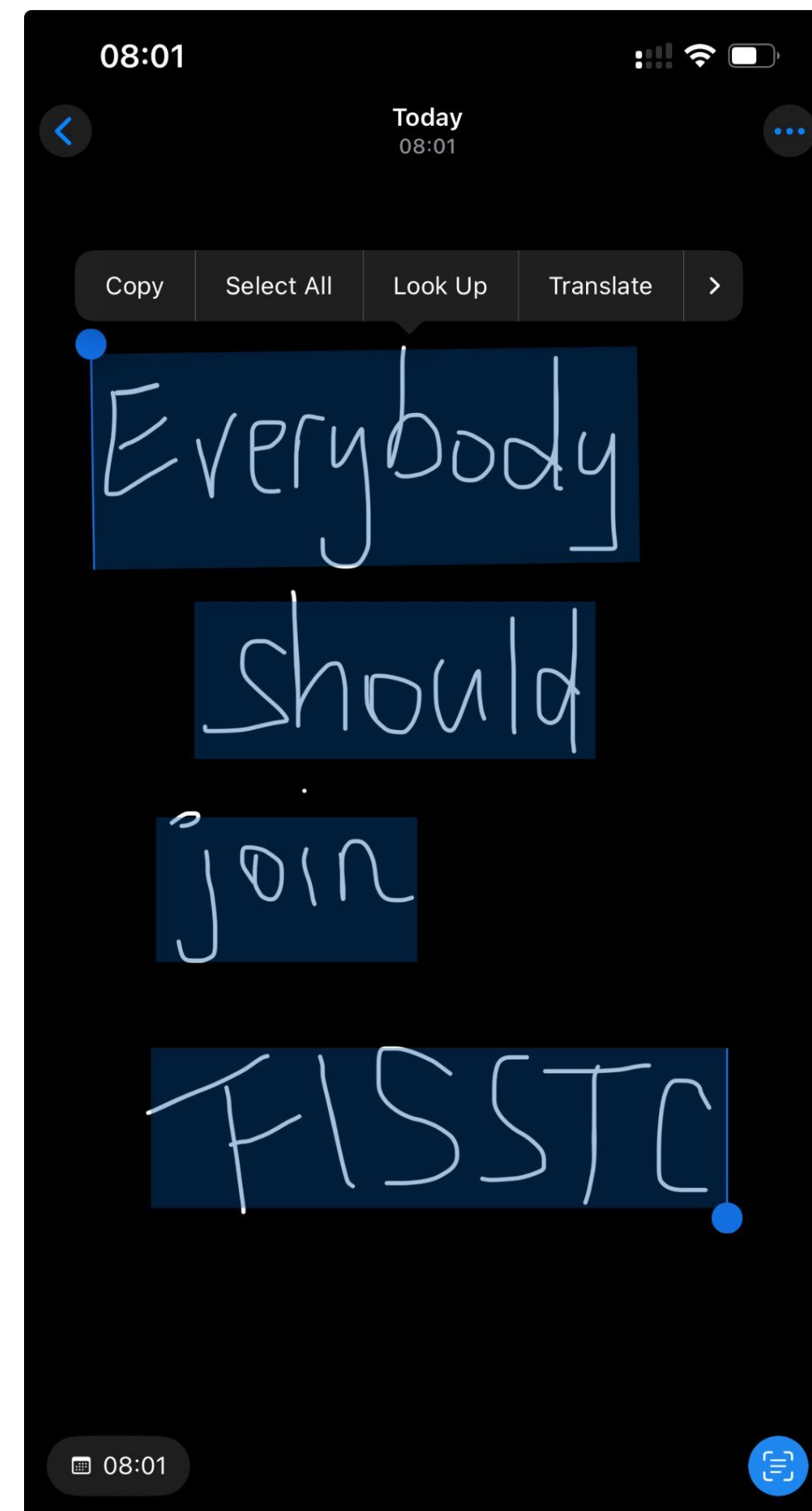
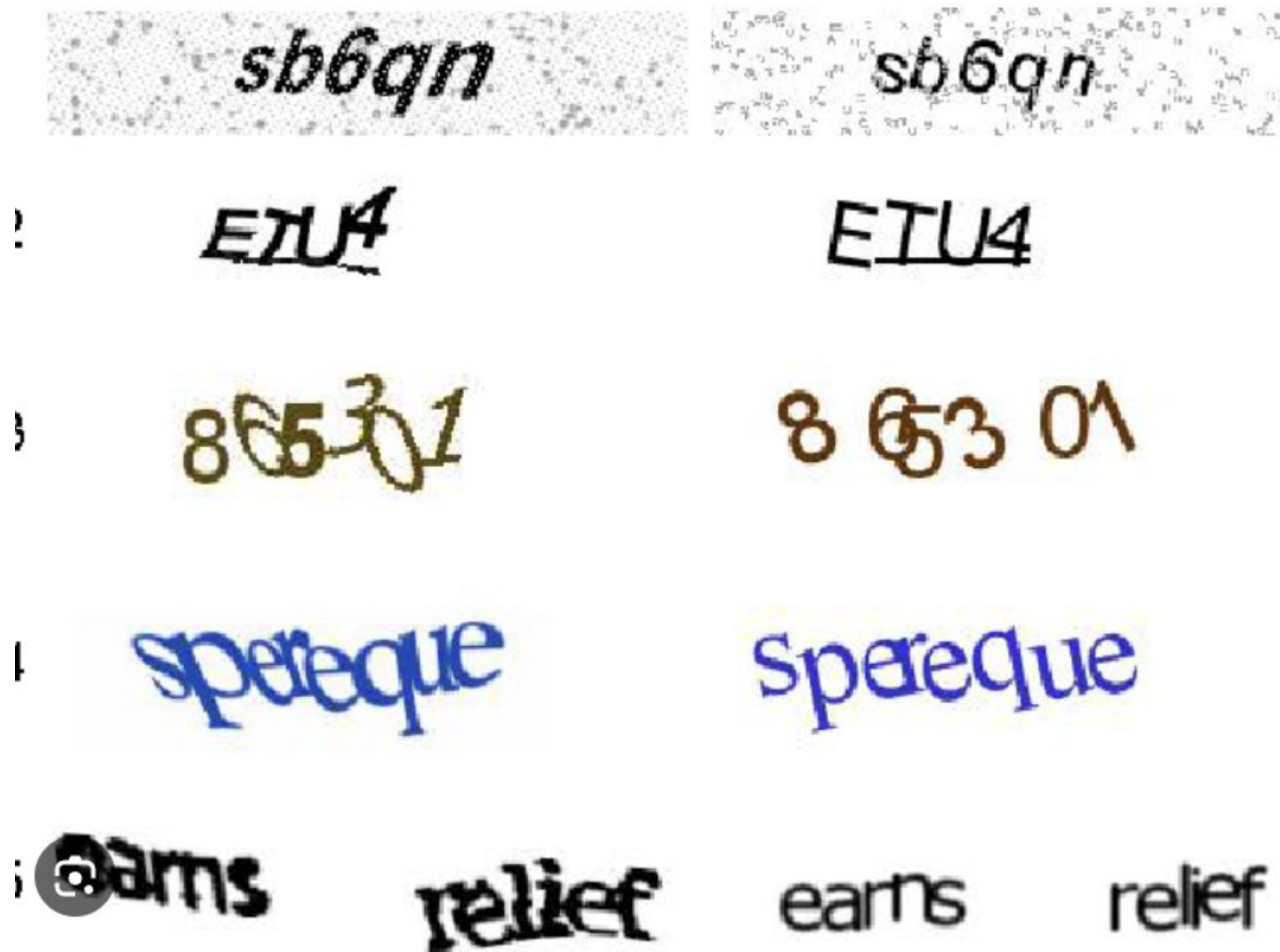


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<https://www.pandasecurity.com/en/mediacenter/what-is-captcha/>



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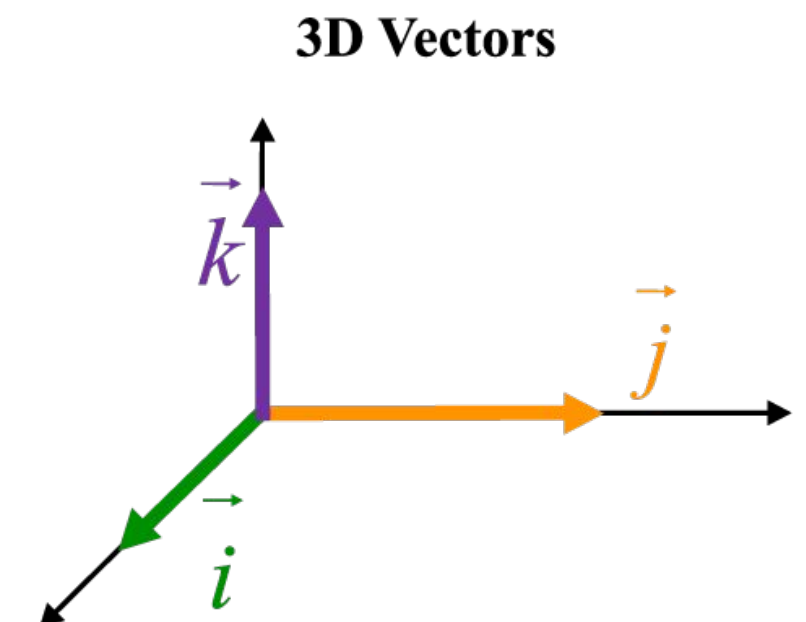
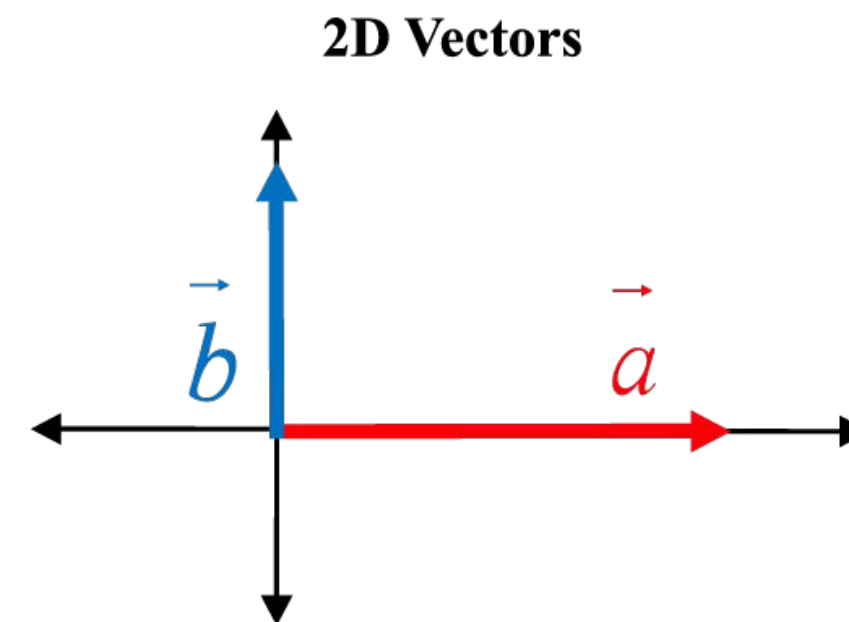
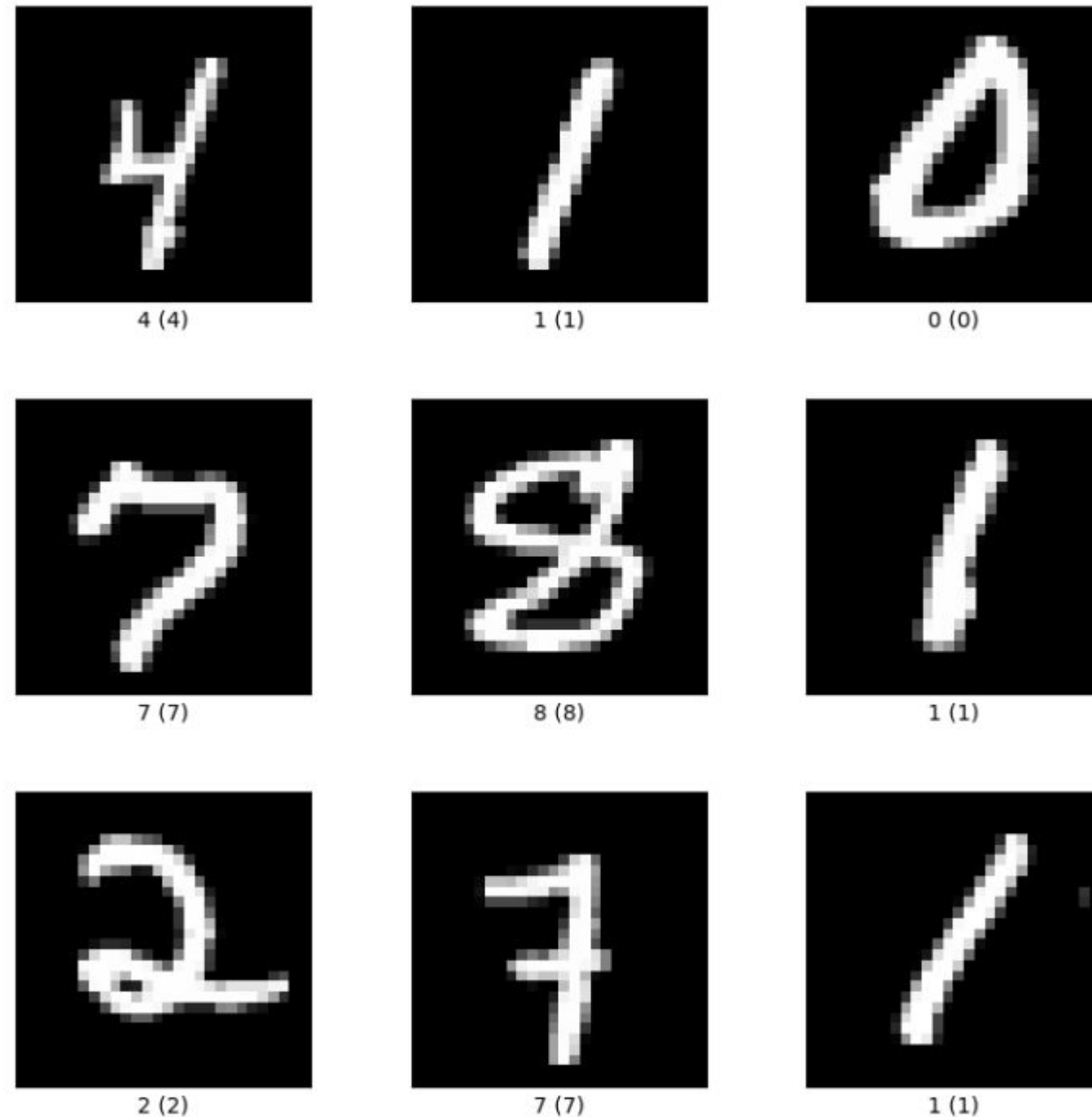
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MNIST and Vectors

(Modified National Institute of Standards and Technology)

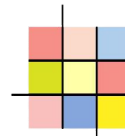


70,000 grayscale images of handwritten digits from 0 to 9, each sized at 28×28 pixels



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PROBLEM STATEMENT

Our study focuses on ...

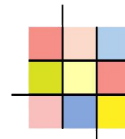
RESEARCH QUESTION

How does ... affect ...



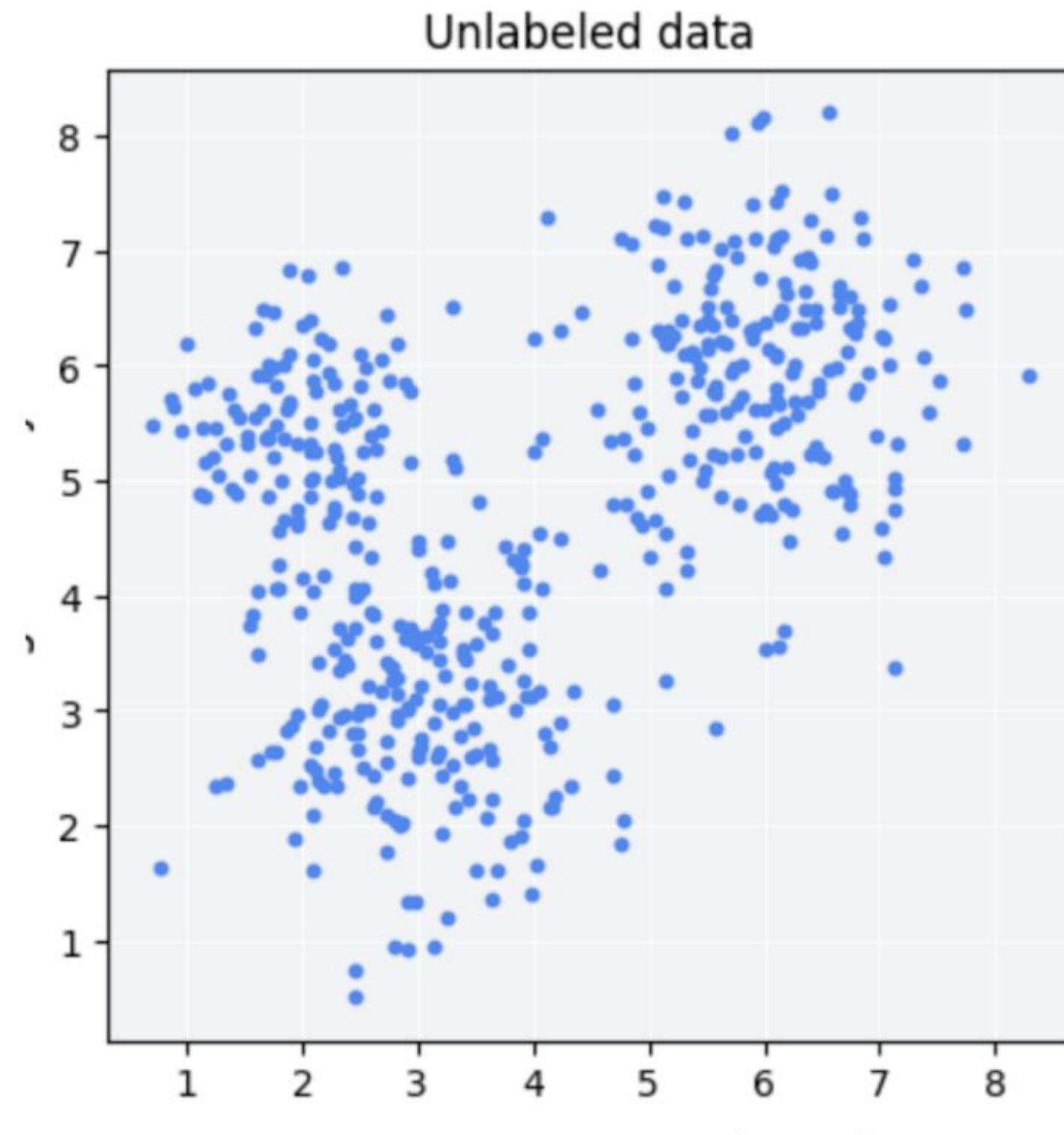
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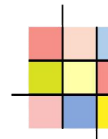
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Clustering



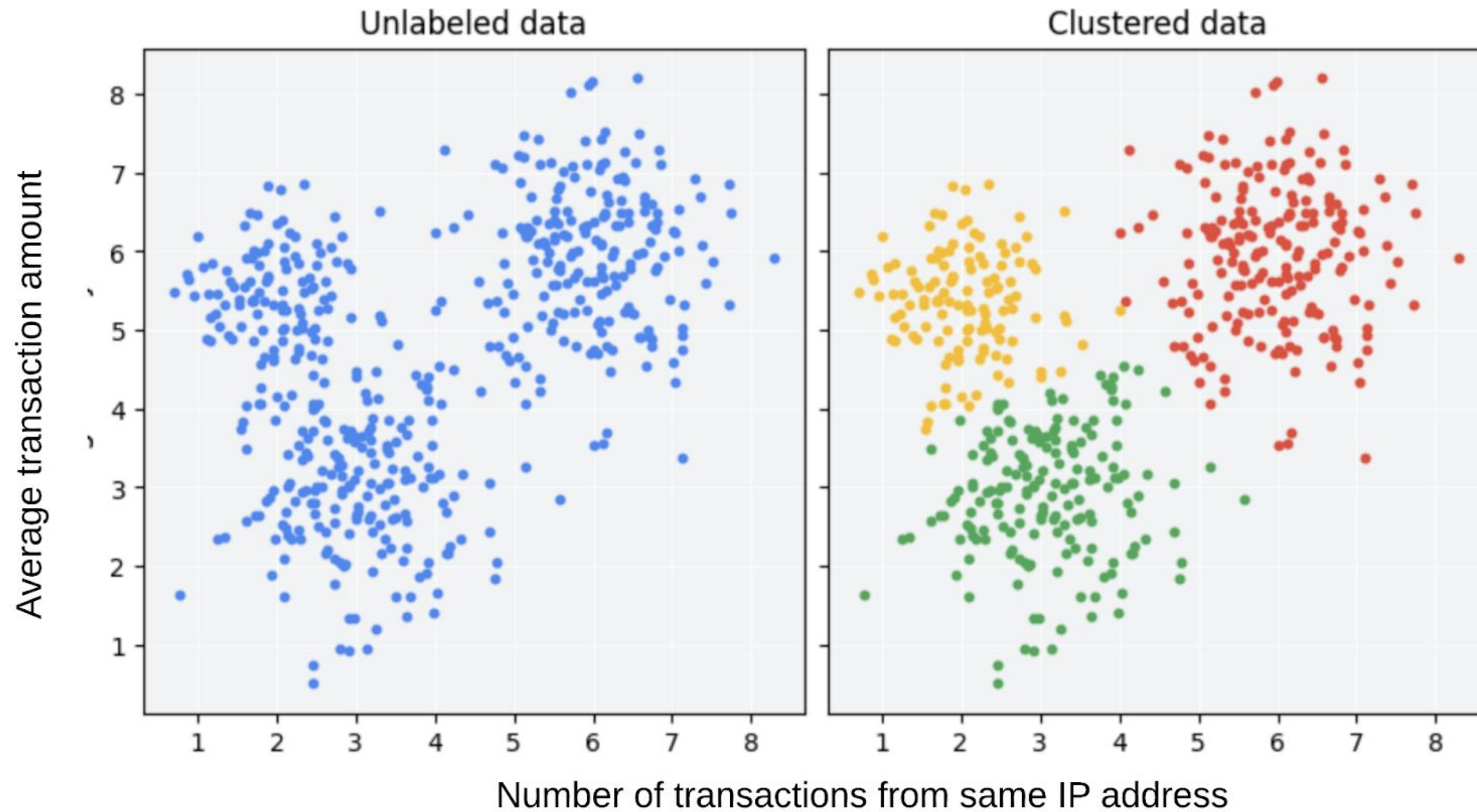
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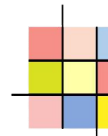
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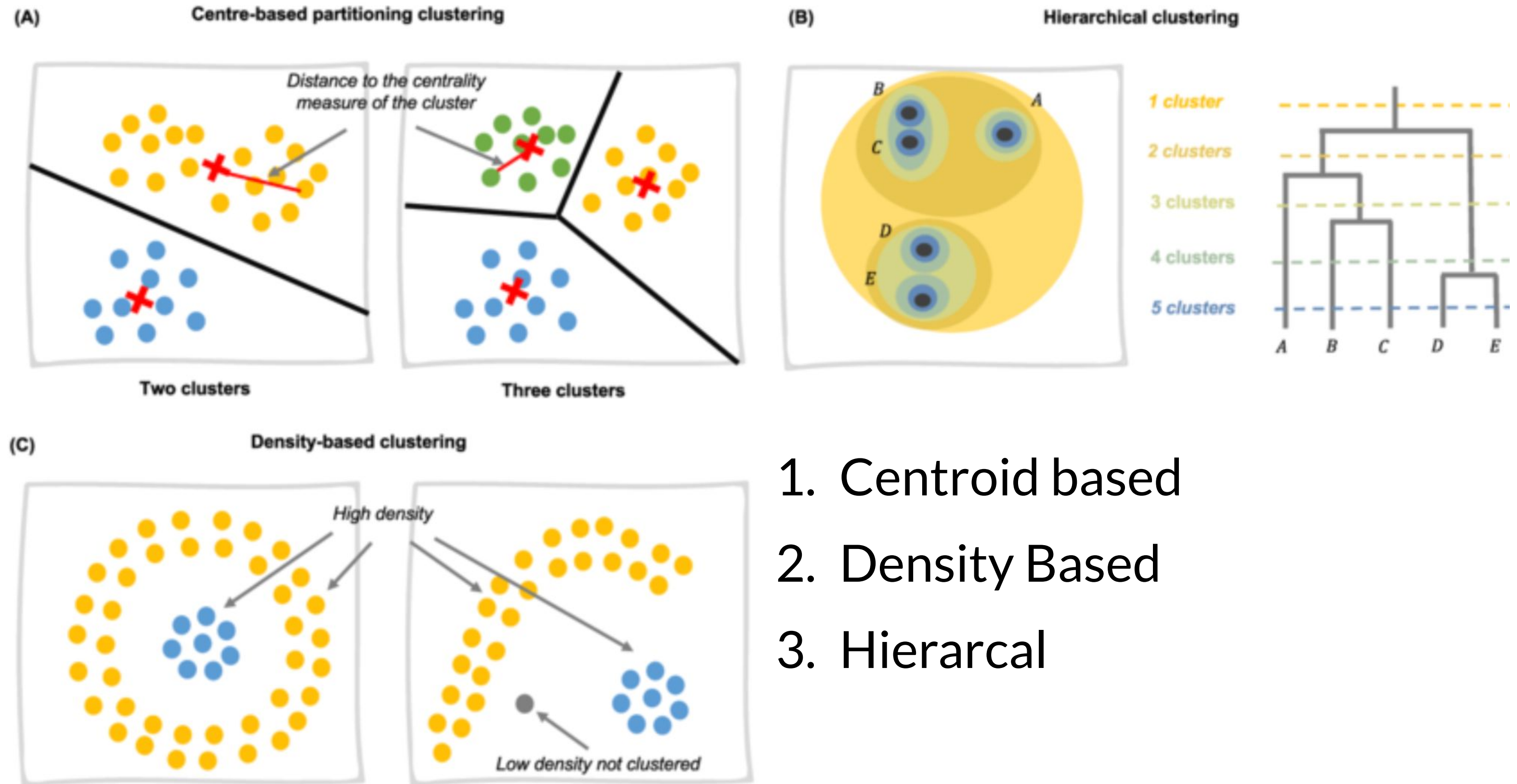
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Clustering Algorithms



1. Centroid based
2. Density Based
3. Hierarcal



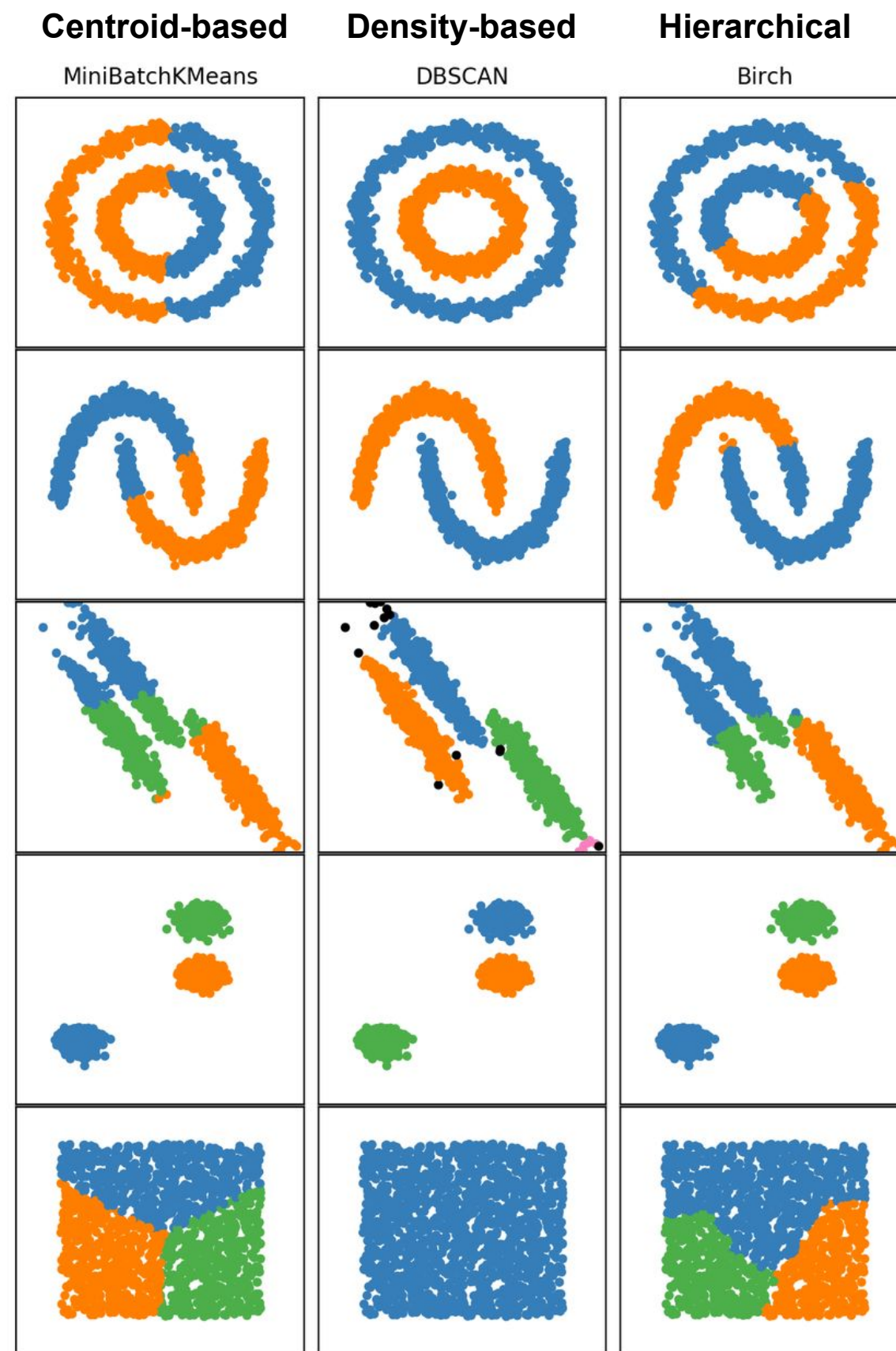
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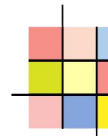
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Clustering Algorithms



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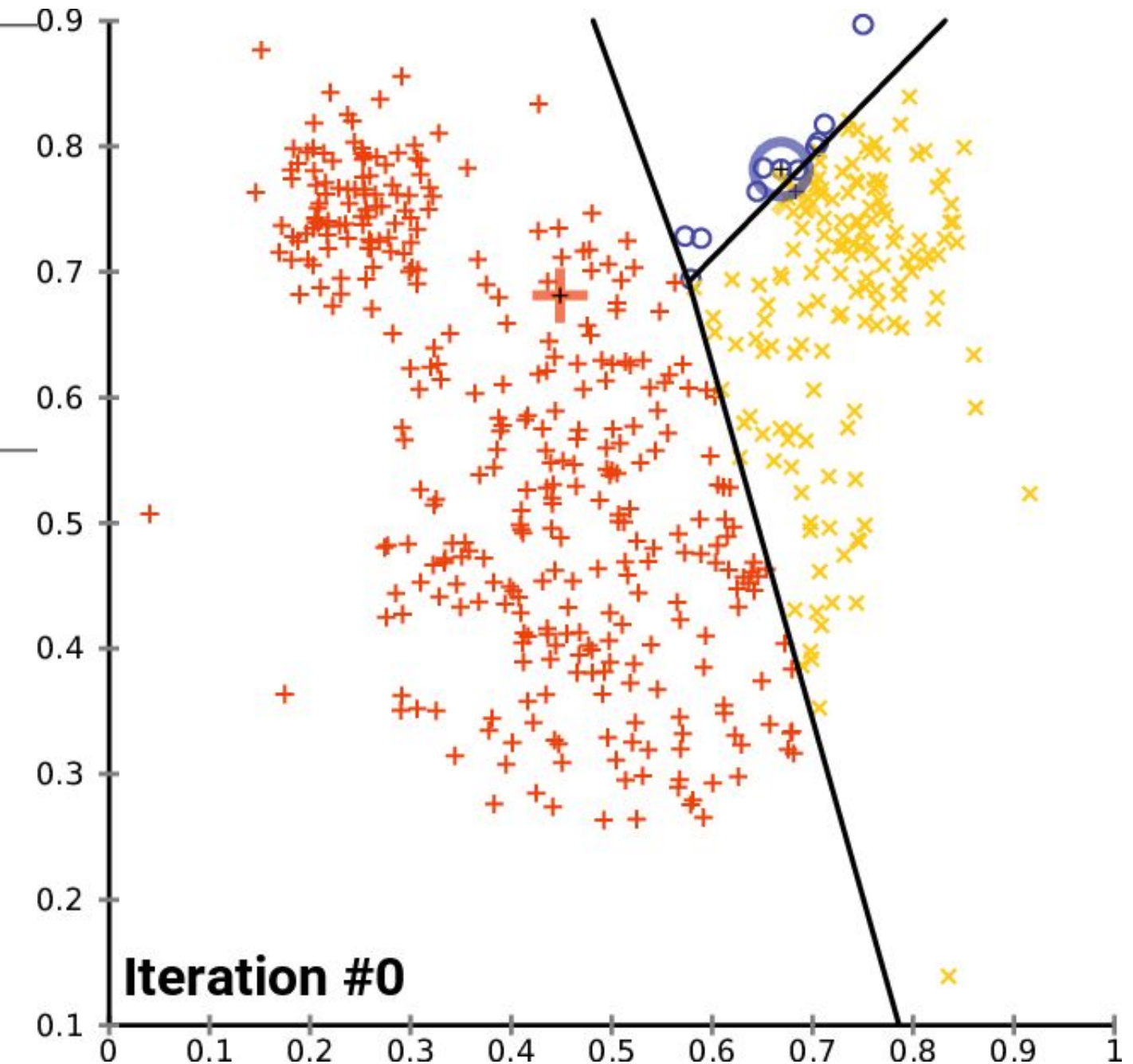
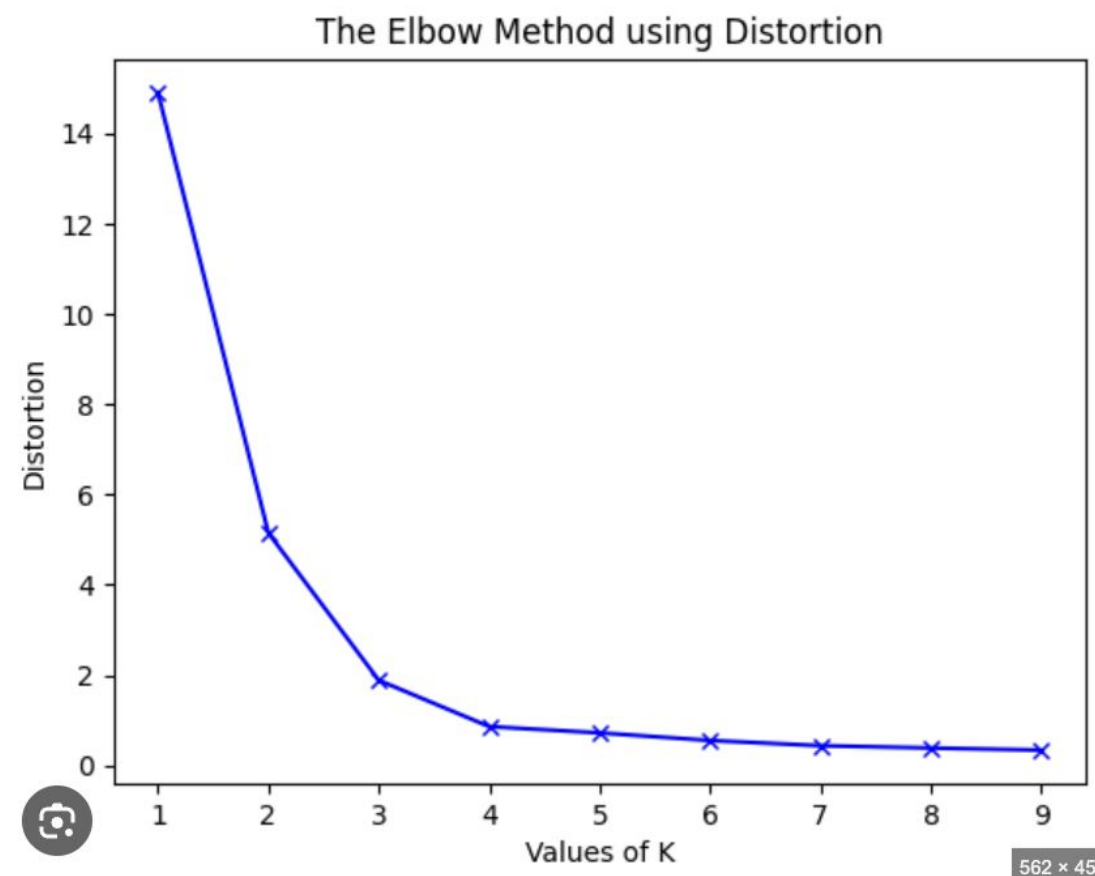


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K-means

Algorithm 1 k -means algorithm ((Arvai))

- 1: Specify the number k of clusters to assign.
 - 2: Randomly initialize k centroids.
 - 3: **repeat**
 - 4: **Expectation:** Assign each point to its closest centroid.
 - 5: **Maximization:** Compute the new centroid (mean) of each cluster.
 - 6: **until** The centroid positions do not change.
-



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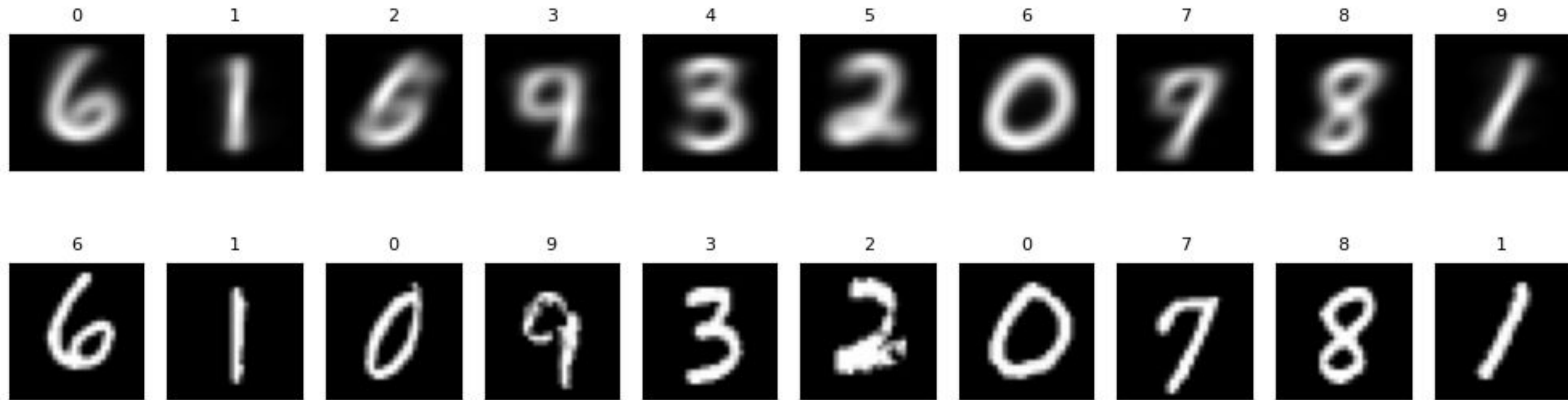
1st Experiment

Algorithm 1 <i>k</i> -means algorithm ((Arvai))	
1:	Specify the number <i>k</i> of clusters to assign.
2:	Randomly initialize <i>k</i> centroids.
3:	repeat
4:	Expectation: Assign each point to its closest centroid.
5:	Maximization: Compute the new centroid (mean) of each cluster.
6:	until The centroid positions do not change.

Table 1: Parameters tested in Experiment 1

Parameter	Description	Values Tested
n_clusters	Number of clusters	10, 25, 50, 75, 100
n_init	Number of centroid initializations	10, 20, 50
max_iter	Maximum iterations per cluster run	100, 300, 500

Results



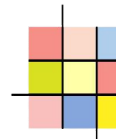
Time (s): 12.43

Params: n_clusters=25, n_init=50,
max_iter=300



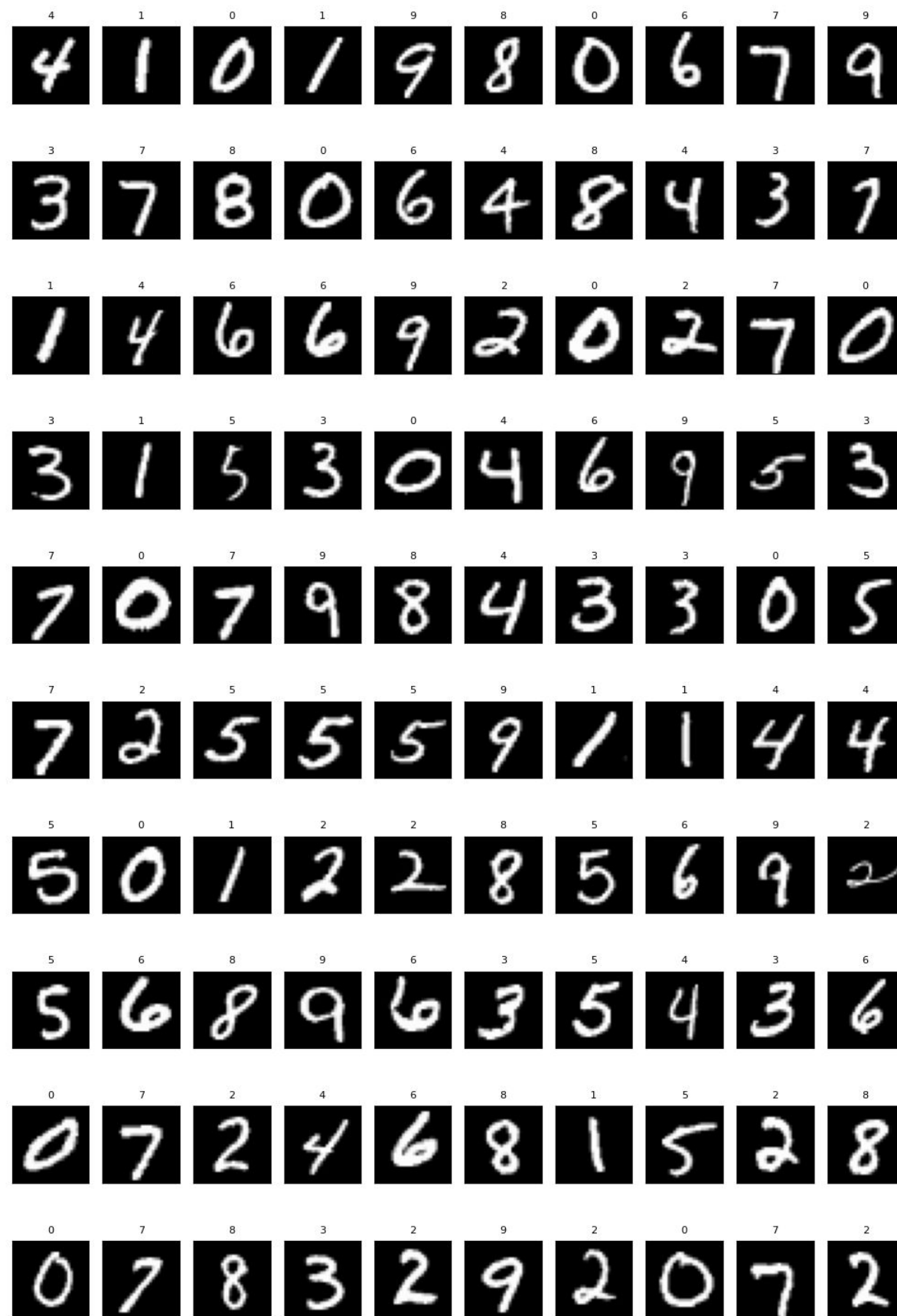
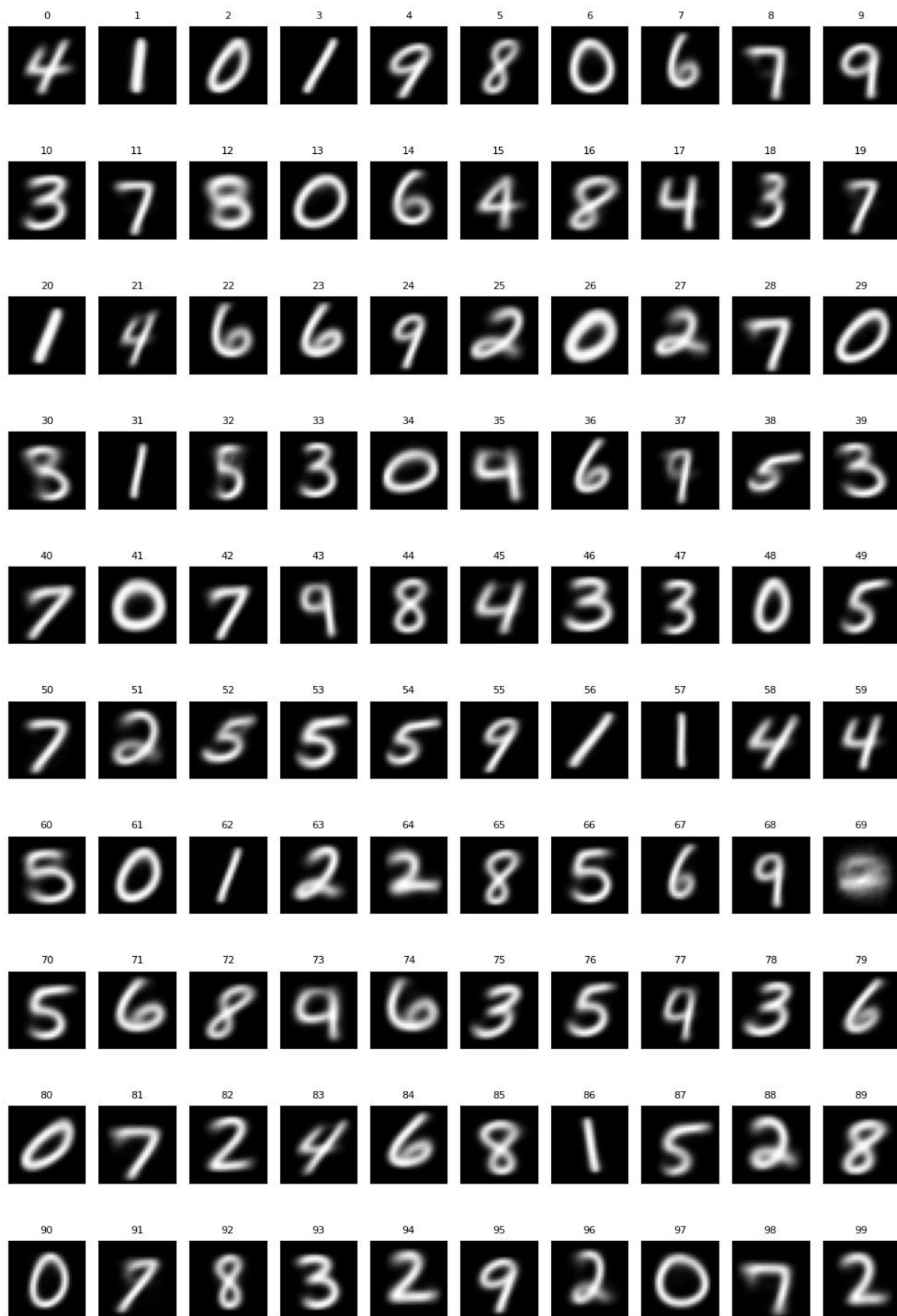
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Results



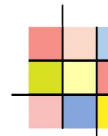
Time (s): 176.20

Params: n_clusters=100, n_init=50,
max_iter=300



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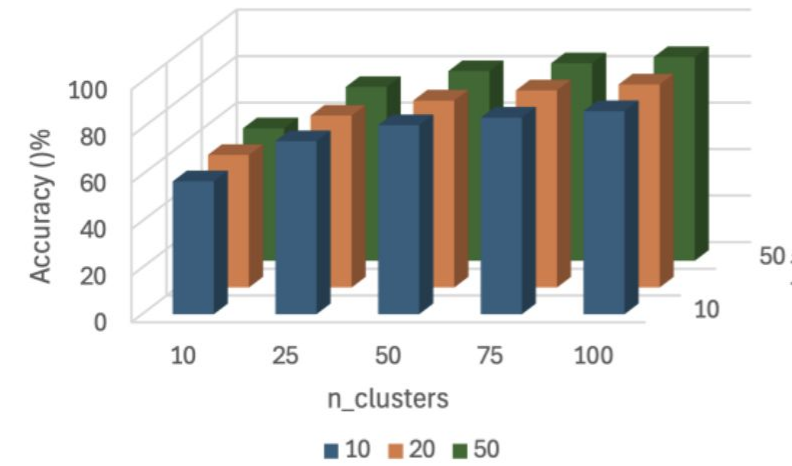


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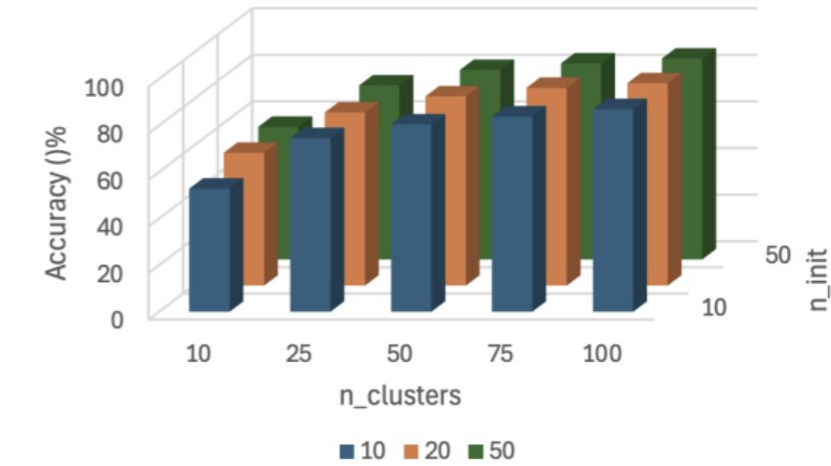
Results

Table 3: Experiment 1 Results – Time and Accuracy across Trials

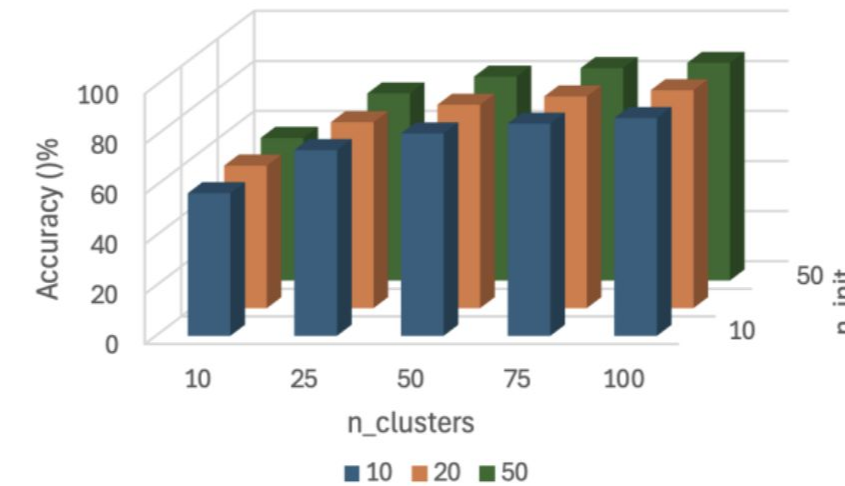
max_iter	n_init	n_clusters	TIME (s)			ACCURACY (%)		
			Trial 1	Trial 2	AVG	Trial 1	Trial 2	AVG
100	10	10	7.2	7.4	7.3	57.1	57.2	57.1
100	10	25	12.4	12.4	12.4	74.1	74.7	74.4
100	10	50	19.5	18.5	19.0	81.5	81.1	81.3
100	10	75	25.3	25.4	25.4	84.5	84.6	84.5
100	10	100	31.4	31.5	31.5	87.2	87.5	87.4
100	20	10	13.9	14.1	14.0	57.0	57.1	57.0
100	20	25	24.6	23.9	24.2	74.0	73.9	74.0
100	20	50	36.2	37.3	36.8	80.6	80.2	80.4
100	20	75	52.1	50.2	51.1	85.0	84.6	84.8
100	20	100	65.6	64.1	64.9	87.8	86.8	87.3
100	50	10	34.7	32.9	33.8	57.0	57.0	57.0
100	50	25	59.9	59.7	59.8	74.7	74.8	74.8
100	50	50	96.4	95.2	95.8	81.6	81.6	81.6
100	50	75	127.8	129.0	128.4	84.8	85.0	84.9
100	50	100	155.1	156.2	155.7	87.4	87.8	87.6
300	10	10	8.4	6.4	7.4	52.9	52.9	52.9
300	10	25	13.8	13.1	13.4	74.9	74.3	74.6
300	10	50	23.2	22.1	22.6	80.0	81.0	80.5
300	10	75	29.0	31.0	30.0	84.1	83.7	83.9
300	10	100	38.0	35.7	36.9	87.0	87.1	87.0
300	20	10	15.3	15.2	15.3	57.0	57.2	57.1
300	20	25	25.7	29.0	27.4	74.8	73.9	74.3
300	20	50	45.2	44.7	45.0	81.3	81.3	81.3
300	20	75	56.7	57.5	57.1	86.1	83.9	85.0
300	20	100	67.8	72.5	70.1	87.5	86.4	87.0



(a) graph of n_cluster and n_init against accuracy, with max_iter of 100



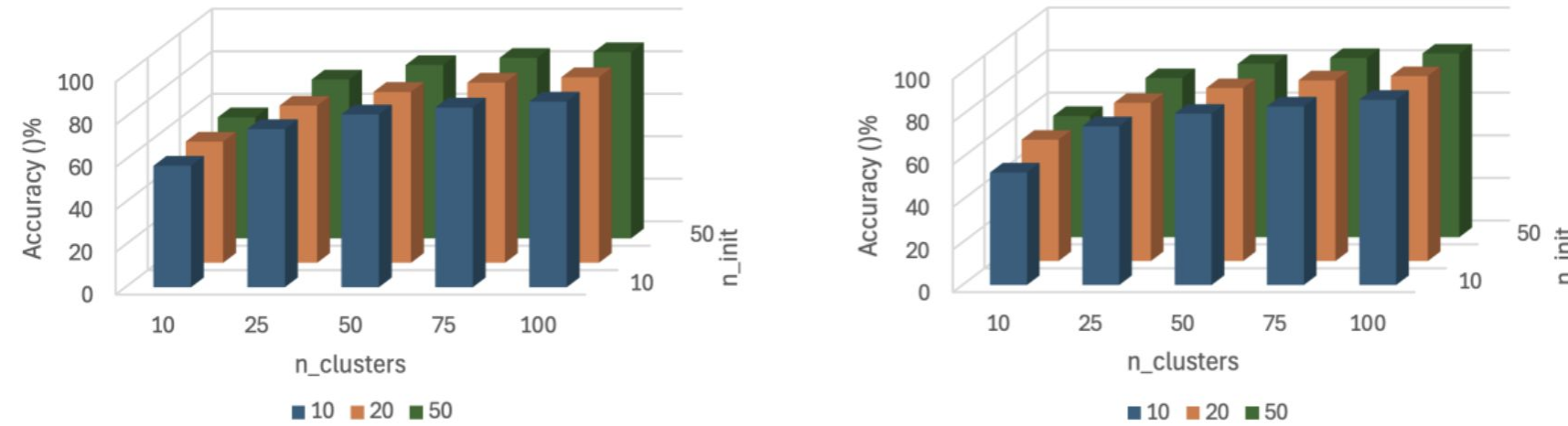
(b) graph of n_cluster and n_init against accuracy, with max_iter of 300



(c) graph of n_cluster and n_init against accuracy, with max_iter of 500

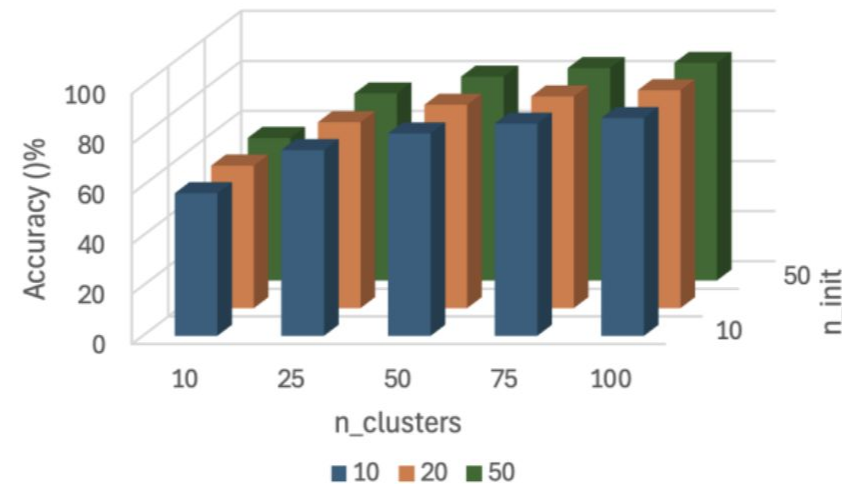
Figure 2: Processed accuracy graphs across parameter settings.

Results



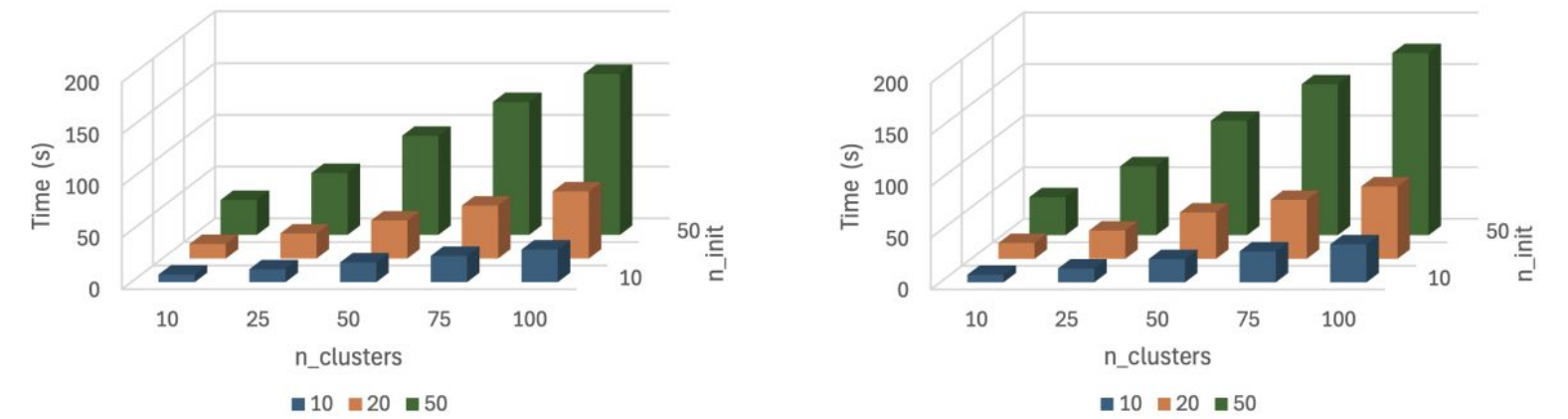
(a) graph of n_cluster and n_init against accuracy, with max_iter of 100

(b) graph of n_cluster and n_init against accuracy, with max_iter of 300



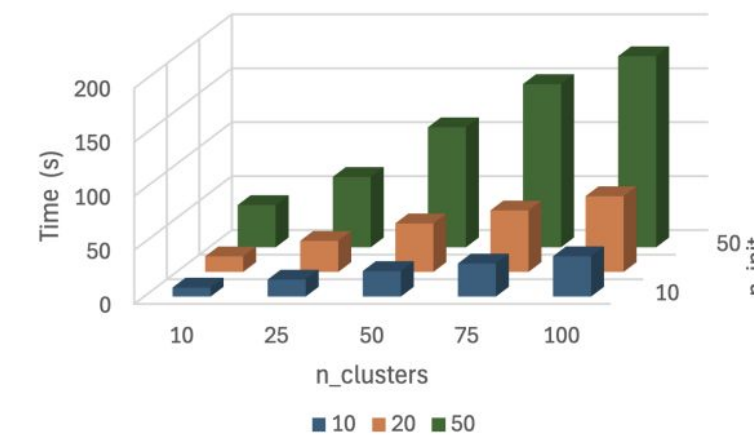
(c) graph of n_cluster and n_init against accuracy, with max_iter of 500

Figure 2: Processed accuracy graphs across parameter settings.



(a) graph of n_cluster and n_init against time, with max_iter of 100

(b) graph of n_cluster and n_init against time, with max_iter of 300



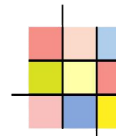
(c) graph of n_cluster and n_init against time, with max_iter of 500

Figure 3: Processed time graphs across parameter settings.



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2nd Experiment

Table 1: Parameters tested in Experiment 1

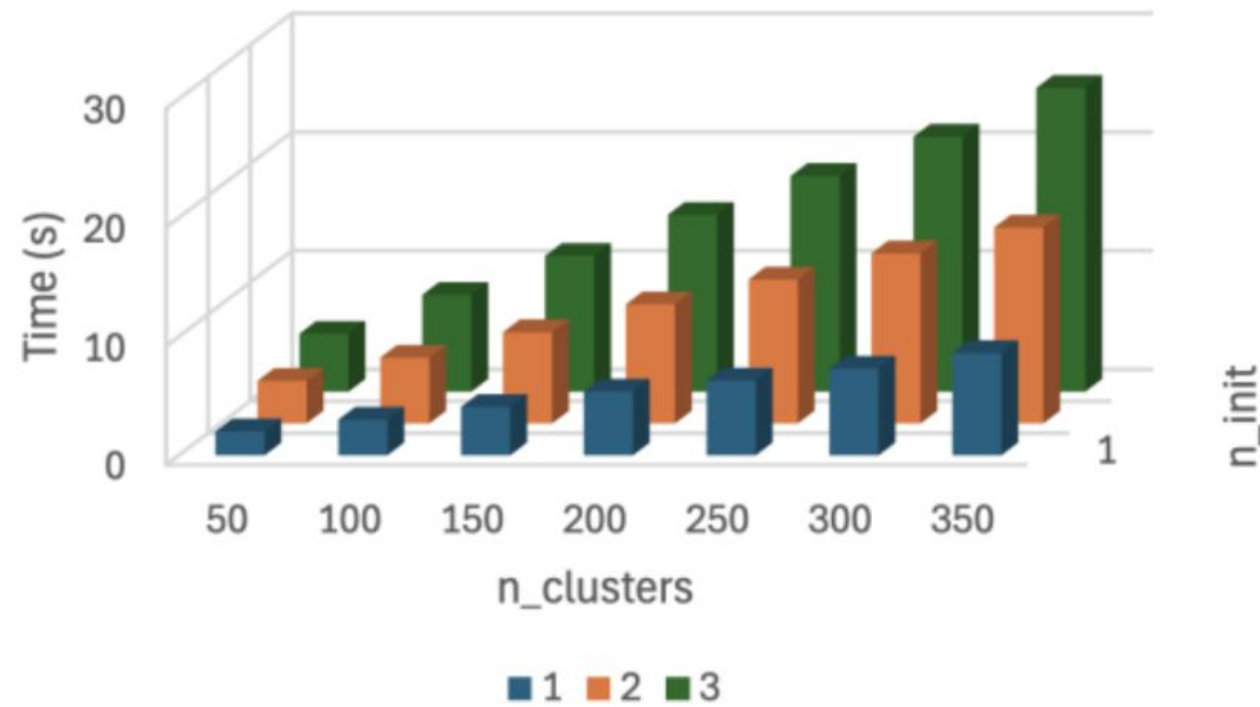
Parameter	Description	Values Tested
n_clusters	Number of clusters	10, 25, 50, 75, 100
n_init	Number of centroid initializations	10, 20, 50
max_iter	Maximum iterations per cluster run	100, 300, 500

Table 2: Parameters tested in Experiment 2

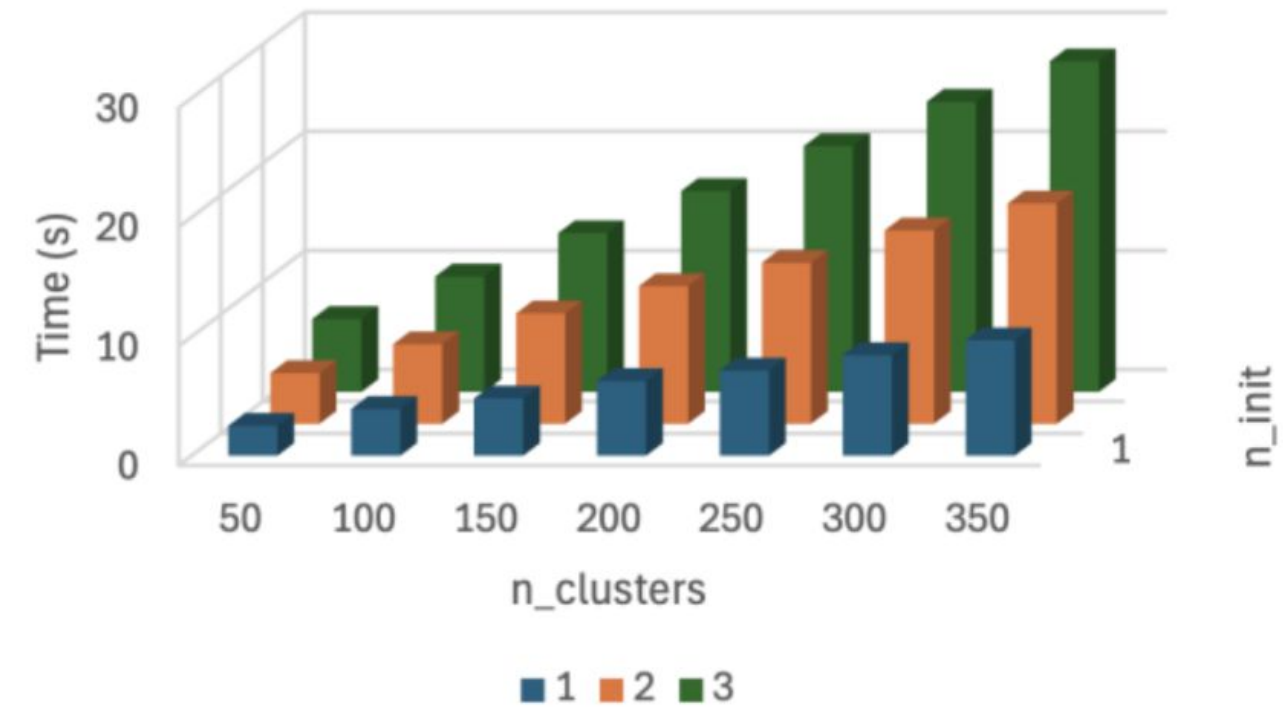
Parameter	Description	Values Tested
n_clusters	Number of clusters	50, 100, 150, 200, 250, 300, 350
n_init	Number of centroid initializations	1, 2, 3
max_iter	Maximum iterations per cluster run	50, 100



Results



(a) graph of n_cluster and n_init against time, with max_iter of 50



(b) graph of n_cluster and n_init against time, with max_iter of 100

Figure 6: Processed time graphs across parameter settings.



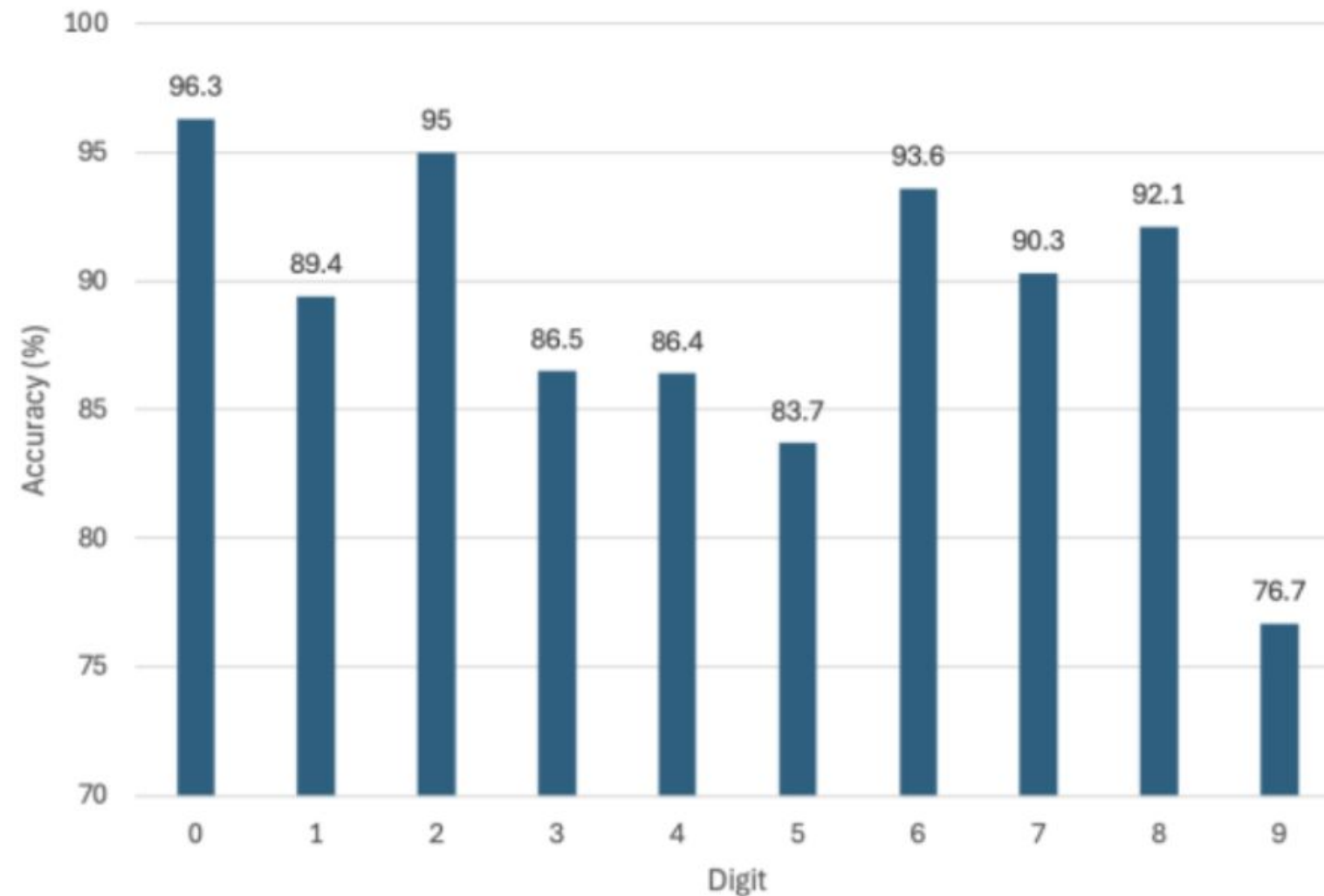
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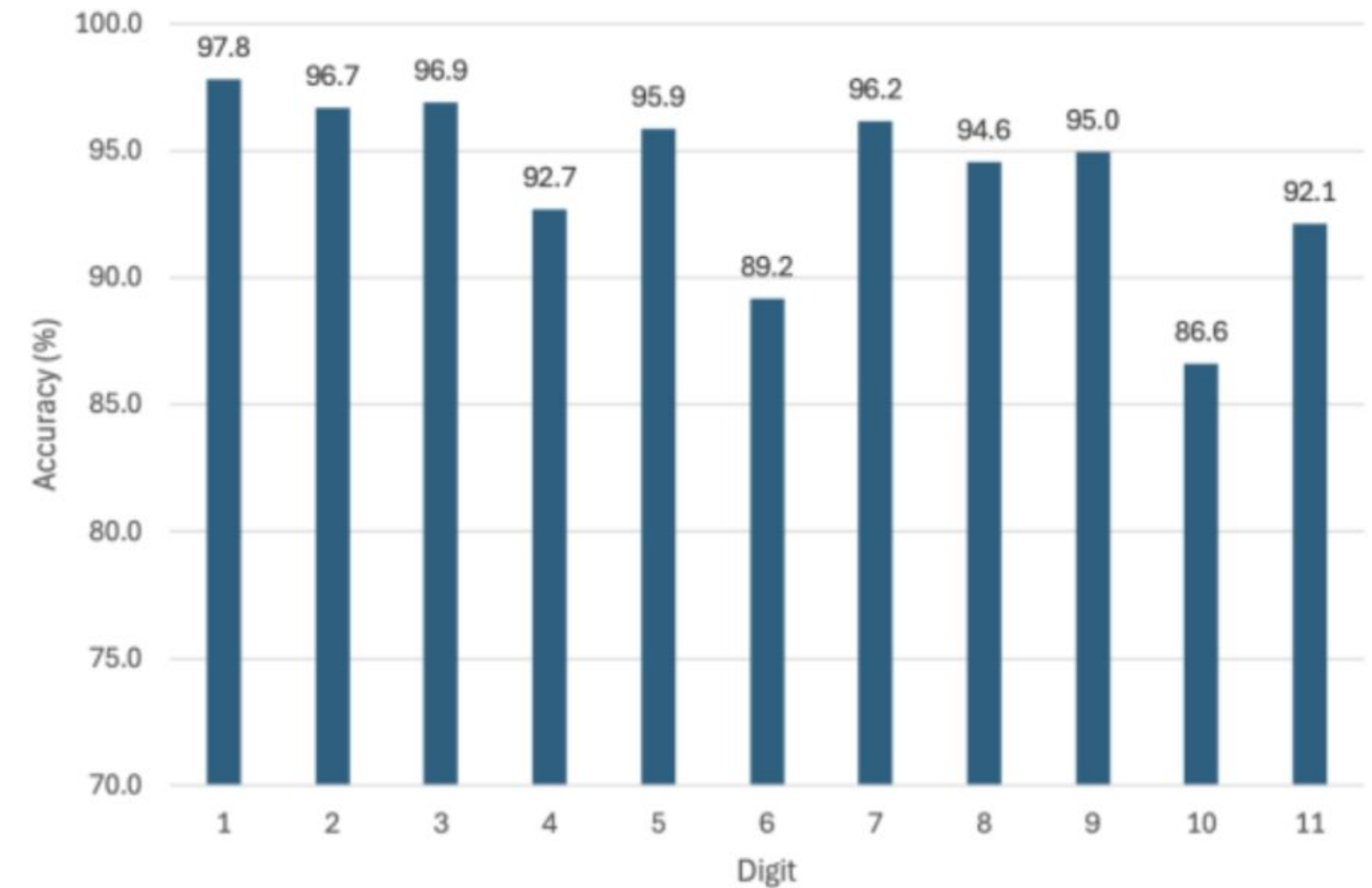


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Results



(a) Average Accuracy of each digit



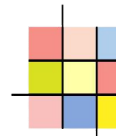
(b) Maximum Accuracy of each digit

Figure 7: Accuracy of each digit achieved during Experiment 2



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Clustering

Thank you for your attention

Any Questions?

Zoe